

Numerical modeling of microbubble backscatter to optimize ultrasound particle image velocimetry imaging: initial studies

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Abstract

We have developed a promising non-invasive ultrasound-based method for performing particle image velocimetry (PIV) in vivo. This method, termed echo PIV, provides multi-component blood velocity data with good (~ 2 ms) temporal resolution. The method takes advantage of the non-linear ultrasound backscatter characteristics of small gas-filled microbubbles (ultrasound contrast) that are seeded into the blood stream. In this study, we use a numerical model to explore potential areas to focus future work in echo PIV.

Ultrasound backscatter from encapsulated microbubbles was modeled using a modified Rayleigh–Plesset equation (Church model, 1995), taking into account the protein/lipid shell layer as a thick, mass-conserving incompressible fluid surrounded by incompressible blood-like fluid. The equation of motion was solved numerically to characterize the fundamental and second harmonic components of the backscattered pressure. Results show a significant advantage in using the second harmonic component for echo PIV, especially for small bubble sizes less than $3\text{ }\mu\text{m}$ in diameter at 2.2 MHz frequency. The effect of the shell thickness ranging from 10 to 500 nm on the vibration amplitude of the bubble was examined and it is shown that the presence of the shell requires mechanical index (MI) > 0.2 of incident pressure amplitude to improve bubble detectability. Analysis of the effect of pulse length shows a tradeoff between axial resolution (short pulse length) and bubble detectability (longer pulse length) will most likely be required. The effect of varying MI between 0.1 and 0.6 was also studied at a center frequency of 2.2 MHz and the results indicate that the resonance of the second harmonic is maximized for bubbles with diameter of approximately $2.75\text{ }\mu\text{m}$. Bubble non-linearities at MI > 0.2 induced a resonant frequency shift away from the integer multiple of the incident frequency in the second harmonic backscatter. For a given bubble size, there is a combination of optimal incident frequency and mechanical index range that maximizes the ratio of the second harmonic compared to the fundamental. This resonant frequency decreases with increasing bubble radius. Further, a narrow bandwidth pulse is shown to increase signal strength. Both of these effects may cause conflict with factors governing spatial resolution. Optimization of the incident frequency, microbubble size and mechanical index to enhance bubble detectability will depend on the particular clinical application. These theoretical predictions provide further understanding of the physics behind our echo PIV technique, and should be useful for guiding the design of echo PIV systems.

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1. Introduction

We have recently developed a novel non-invasive ultrasound-based method to perform particle image velocimetry in opaque flows such as blood flow in the

body. This method, termed echo PIV, provides multi-component blood flow data with good (~ 2 ms) temporal resolution through the use of ultrasound contrast agents (UCAs), which are small gas-filled microbubbles seeded into the blood stream [1]. The method has shown initial promise in measuring velocity profiles and wall shear in mock arteries and in quantifying diastolic flow in animal models and clinical studies [2,3]. Although this technique has been implemented with current clinical ultrasound systems, a wider range of clinical applications will

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be possible with improved spatial and temporal resolution; maintaining and increasing UCA visibility is also a key component. In this regard, a better understanding of the relationship between the physics of the backscatter process from microbubbles and the requirements for echo PIV is an essential first step to guide the next stage of hardware design. In this paper, we present and discuss initial results from numerical modeling of microbubble backscatter within the context of echo PIV development.

Echo PIV is the *in vivo* equivalent of conventional laser-based particle image velocimetry techniques that have matured in the fluid diagnostics area over the last decade. We will briefly outline the requirements for PIV below; we refer the reader to past work for further details [4–6]. Particle image velocimetry follows the motion of particles seeded into the flow field to measure the local velocity vector in two-dimensions. In optical PIV, particles that scatter light strongly such as silver-coated glass beads are used; a cross-sectional plane of the flow field is illuminated using a thin sheet of light, and scattered light signals from individual particles moving within this plane are recorded by a digital camera. The light sheet is pulsed at very short time intervals and consecutive frames of particle positions are captured. In echo PIV, ultrasound B-mode (brightness proportional to backscattered intensity) images of UCA particles are obtained at high frame rates. As in optical PIV, two sequential images are processed to determine particle displacements, and hence the velocity field.

PIV processing consists of FFT-based cross-correlation between small (8×8 pixel minimum) sub-windows from the two images. This process is repeated over the entire image frame to create a map of the velocity vector field. Advanced signal-processing techniques are usually applied to increase the spatial resolution. Windows are usually overlapped by 50%. In addition, the second interrogation window is moved to the expected position of the matching particles [7], and an iterative scheme to improve the interrogation window size is usually used [8,9]. The accuracy of the PIV algorithm depends on a number of variables, but the ability to detect individual particles distinctly from background noise and near the vicinity of solid structures is a key requirement since this directly impacts the cross-correlation results. This requirement underlies the purpose behind this study.

Although it is well known that UCA microbubbles exhibit non-linear behavior, details of this process need to be better elucidated as they relate to this quantitative velocimetry technique. For example, we applied PIV algorithms directly to B-mode images using backscatter at the fundamental frequency to generate velocity maps and tested this technique in mock arteries [3]. We were able to obtain multi-component velocity vectors and instantaneous flow rate with reasonable accuracy. However, this study also revealed error in estimation of

dynamic shear in pulsating flow conditions. There were two reasons for this: insufficient spatial resolution in regions of high velocity gradients, and problems with imaging microbubble signatures in the vicinity of a solid structure. The first problem can be addressed using a combination of high frequency transducers, since spatial resolution is a direct function of excitation frequency, and linear arrays to improve uniformity of the acoustic field and lateral resolution. The second problem is more fundamental since the blood-tissue interface will always produce strong backscatter, which makes it difficult to isolate backscatter from microbubbles in the proximity of such interfaces even with the enhanced scatter produced by such bubbles. This points to the importance of being able to discriminate bubble backscatter from tissue backscatter, and provides the context for our examination of backscatter physics, the initial results from which are reported here.

2. Prior studies of microbubble dynamics

The study of acoustic excitation of bubbles has a long history. Last century, Lord Rayleigh [10] derived the basic governing equation of radial motion of gas-filled bubbles. Later, the bubble equation of motion was modified to take surface tension and fluid viscosity into account [11–13]. This is the well-characterized Rayleigh–Plesset equation. Extensive investigation on the subject of non-linear bubble dynamics has been conducted including application of modern theories of non-linear dynamics to the Rayleigh–Plesset equation [11–17]. However, investigations examining non-linear coated-microbubble dynamics in regimes appropriate for echo PIV optimization have received less attention.

Bubble dynamics and backscatter signal generation are both dependent on incident acoustic pulse amplitude and frequency. The mechanical index ($MI = P_a / \sqrt{f_c}$ where P_a = peak rarefactional amplitude; f_c = center frequency) is used on commercial medical ultrasound systems as a measure of acoustic pressure and potential tissue toxicity. At mechanical index (<0.2), bubble motion can be classified as quasi-linear, oscillating at the driving pulse frequency. On the other hand, increasing mechanical index (>0.2) yields harmonic generation in the backscattered signal as a result of non-linear bubble motion.

Echo PIV uses contrast agents that are encapsulated gas-filled microbubbles, typically 1–10 μm in diameter and coated with a 10–200 nm thick protein or lipid shell [18]. Two agents that have been used clinically are Albunex® (Mallinckrodt Medical, St. Louis, MO) and Optison® (Amersham Health, Princeton, NJ). Several others are being examined for clinical use. The standard Rayleigh–Plesset equation models air bubbles without encapsulation; the coating used in ultrasound contrast

increases mechanical stability but decreases backscattered pressure. Furthermore, encapsulated microbubbles have higher resonant frequencies than free microbubbles [19]. The encapsulating shell, therefore, needs to be taken into account when modeling ultrasound backscatter from contrast agents. Elasticity and viscosity parameters of the shell were first reported by de Jong et al. [20–24] who modified the Rayleigh–Plesset equation of a free bubble using in vitro measurements of coating-layer elasticity and damping. These authors studied the effect of large amplitude incident pressures on radial deformation and resulting non-linear backscatter, and concluded that significant harmonic generation requires excitation pulse frequency (>1 MHz) and pressure amplitude (>200 kPa). In order to accurately determine coating-layer elastic and viscous parameters within high concentrations of contrast, bubble interactions also need to be taken into account. Such interactions, specifically dispersion and attenuation of waves propagating in a chamber filled with different concentrations of contrast (Albunex[®]), have also been studied [25–29]. The latter authors found different shell properties, i.e., elasticity and viscosity, than those reported earlier by de Jong et al. [20–24]. The differences in shell properties can be explained most likely by the different concentrations of microbubbles used in these experiments.

Equations describing the dynamics of encapsulated microbubbles have been examined by Church [30] who derived the equations of radial bubble motion taking into account the shell properties, i.e., the shell mass, stiffness, and viscosity. Church [30] derived a Rayleigh–Plesset-like equation using linearly visco-elastic constitutive equations for the shell, which is valid for bubble deformation with oscillation less than 100% of the initial diameter [31]. Another approach presented by Allen et al. [32] considered the shell as a thick, mass-conserving incompressible fluid. They studied the radial deformation of oscillating microbubbles without characterizing the scattered pressure signals.

For practical application of our echo PIV technique, however, investigation of separate harmonic components in the backscattered signal is required since signals at these frequencies may provide the means of distinguishing bubble signatures from tissue signals. This is due to the higher level of excitability exhibited by these bubbles, especially in the non-linear range, while tissue response remains primarily quasi-linear. Specifically, the components corresponding to the fundamental and second harmonic need to be examined since many medical ultrasound systems now have the capability to extract the second harmonic component from the backscattered signal. This technique is called second harmonic contrast imaging and is designed to exploit the modal resonance of the bubble and thereby enhance blood flow images [33]. The method extracts and displays signals corresponding to twice the incident fre-

quency from the backscattered signal. The result is that scattering from tissues (native harmonic imaging) has lower amplitudes than second harmonic contrast imaging.

This study provides a theoretical analysis of the backscattered ultrasound pressure within the fundamental and second harmonic frequency components. The bubble is modeled as gas-filled and encapsulated by a thick-walled incompressible viscous shell surrounded by incompressible blood-like fluid. This is analogous to Church [30] model, as modified by Allen et al. [32]. Effects of bubble size and incident signal mechanical index on fundamental and second harmonic backscatter components are examined to determine optimal conditions that will enhance the detection of bubbles for the echo PIV method.

3. Theoretical background

Encapsulated contrast agents are here modeled as spherical air bubbles surrounded by a layer of albumin/lipid shell material, free of body forces. Scattering from surface modes are assumed negligible, and are not considered here. The surrounding fluid is blood-like modeled as a Newtonian fluid. Both the surrounding blood-like fluid and the mass-conservative shell are considered viscous and incompressible. Fig. 1 shows a schematic plot of the encapsulated microbubble with inner and outer radii of R_1 and R_2 respectively. The equations of bubble motion for an encapsulated microbubble can be obtained by combining the continuity and momentum equations for a radial, incompressible potential flow [30,32]:

$$\begin{aligned} & \rho_s R_1 \ddot{R}_1 \left[1 + \left(\frac{\rho_L - \rho_s}{\rho_s} \right) \frac{R_1}{R_2} \right] \\ & + \rho_s \dot{R}_1^2 \left[\frac{3}{2} + \left(\frac{\rho_L - \rho_s}{\rho_s} \right) \left(\frac{4R_2^3 - R_1^3}{2R_2^3} \right) \frac{R_1}{R_2} \right] \\ & = p_g(R_1, t) - \frac{2\zeta_1}{R_1} - \frac{2\zeta_2}{R_2} - P(t) - P_0 \\ & - 4\mu_s \dot{R}_1 \left[\frac{R_2^3 - R_1^3}{R_2^3 R_1} \right] - 4\mu_L \dot{R}_1 \left[\frac{R_1^2}{R_2^3} \right]. \end{aligned} \quad (1)$$

Applying conservation of mass through the shell, the instantaneous outer bubble radius R_2 is related to the inner radius R_1 by

$$R_2^3 = R_{20}^3 - R_{10}^3 + R_1^3, \quad (2)$$

where R_{10} and R_{20} are the equilibrium inner and outer radii of the bubble in static condition.

Solving Eqs. (1) and (2) yields the radial motion of a single bubble exposed to an external acoustic driving pressure $P(t)$. A single bubble model is a reasonable first assumption for the echo PIV environment since the method requires far fewer bubbles than are currently

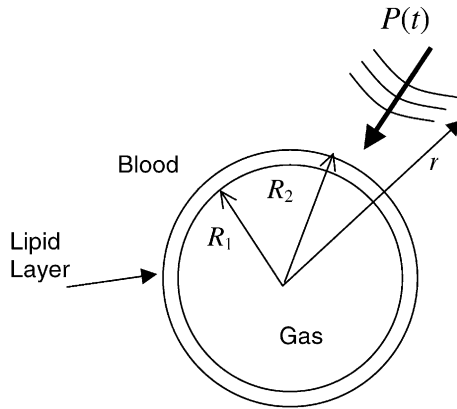


Fig. 1. Schematic plot of encapsulated microbubble filled with gas and surrounded by blood.

used for contrast imaging. In preliminary clinical studies, we have used volumes of 0.001–0.05 cm³ of contrast, followed by a 5 cm³ saline flush, which are several orders of magnitude less than the typical volumes currently indicated for clinical imaging [34]. In our in vitro studies, an optimum concentration of particles was 1 × 10³ particles/ml [1], resulting in a volume fraction of bubbles to liquid of approximately 1 × 10^{−8}.

Numerical solutions of these equations were obtained by implementing a quasi-constant step size technique using implicit Runge–Kutta method [35]. The scattered pressure is obtained from the volume dependence relation [15,17], which can be analyzed in the frequency domain through a fast Fourier transform.

4. Results and discussion

Numerical results presented in this section are for microbubbles coated with a 100 nm lipid shell. The shell is modeled as a thick mass-conserving fluid with mass density $\rho_s = 1.15 \text{ g cm}^{-3}$. The surrounding fluid is modeled as blood, with mass density $\rho_L = 1.0 \text{ g cm}^{-3}$. Viscosity of the shell and blood were $\mu_s = 28 \times 10^{-3} \text{ Pa s}$, $\mu_L = 3 \times 10^{-3} \text{ Pa s}$, respectively. The inner and outer surface tension parameters were taken to be $\xi_1 = 65 \text{ g s}^{-2}$, and $\xi_2 = 5 \text{ g s}^{-2}$, respectively [32]. The acoustic driving pressure incident on the bubble wall was expressed in the form

$$P(t) = P_a \cos(\omega_c t) \exp \left[-\frac{h_d^2 \omega_c^2}{4} (t - t_d)^2 \right], \quad (3)$$

where P_a is the driving pressure amplitude, $\omega_c (= 2\pi f_c)$ the pulse circular center-frequency, h_d the pulse width parameter, t_d the pulse delay time. The mechanical index of the incident pressure is $MI = P_a / \sqrt{f_c}$, with P_a in MPa and f_c in MHz. Mechanical index is a convenient quantity for comparative measurements since it is a parameter that is readily available (and changeable) on current ultrasound imaging systems and since it allows

us to compare modeling results with ongoing experimental studies where MI is also used. However, since MI is a relatively arbitrary parameter, we also provide the pressure amplitude (atm) for comparison.

For imaging techniques, a short pulse (with relatively broadband frequency content) is commonly used to enhance the spatial resolution. In this work, we used a five-cycle harmonic pulse with a Gaussian width envelope. The incident pulse-width parameter is initially taken to be $h_d = 0.2$. Eq. (3) represents a symmetric pulse that does not consider the tissue non-linearity effects, as tissue harmonic generation and pulse waveform distortion. In our preliminary in vitro echo PIV studies we used $MI = 0.4$ ($P_a = 5.93 \text{ atm}$), $f_c = 2.2 \text{ MHz}$; these values, along with typical bubble diameters for ultrasound contrast, were used as a starting point in these numerical studies as well.

Fig. 2 shows the transient radial motion, wall velocities and backscattered pressure for a single microbubble of $R_{10} = 1.5 \mu\text{m}$ inner radius, and 100 nm shell thickness. In the first cycle where the pressure amplitude $P < 5 \text{ atm}$, the bubble oscillates with small radial deformation and small wall velocity. As the wave amplitude increases, the inner radius exhibits sharp collapse and rebound with wall velocities of -110 m/s during collapse and 32 m/s during rebound. However, the outer radius collapses with a wall velocity of -61 m/s and rebounds at 8 m/s . This reduction in wall velocity reflects the damping effect of the coating layer on the dynamic response of the microbubble. Fig. 3 shows the dynamic response of the bubble as mechanical index is increased ($MI = 0.6$, $P_a = 8.90 \text{ atm}$). Significant non-linear behavior of the bubble is now observed with very sharp collapse and rebound; wall velocities of -330 m/s (collapse) and 197 m/s for the inner radius, and -97 and 12 m/s for the outer radius. A modest increase in the mechanical index thus produces dramatic non-linear behavior of the bubble with significant wall velocities and strong resultant backscattered pressure.

The non-linear behavior exhibited by the microbubbles can be better visualized through the use of phase plots, as shown in Figs. 4 and 5 for mechanical index $MI = 0.4$ ($P_a = 5.93 \text{ atm}$), and $MI = 0.6$ ($P_a = 8.90 \text{ atm}$), respectively. These phase plots show wall velocity vs. radial deformation for both inner and outer radii. The bubble radius begins to oscillate around the ambient radius as the pressure amplitude increases. As the incident pressure amplitude increases further, the bubble motion shows strong collapse and a very sudden rebound, with the maximum outward acceleration occurring at minimum radius. This is a result of the large difference in acoustic impedance between the gas inside the microbubble (400 rayl) and the surrounding blood (1.61 Mrayl).

Comparisons between the inner and outer radii behaviors show that the coating produces a damping

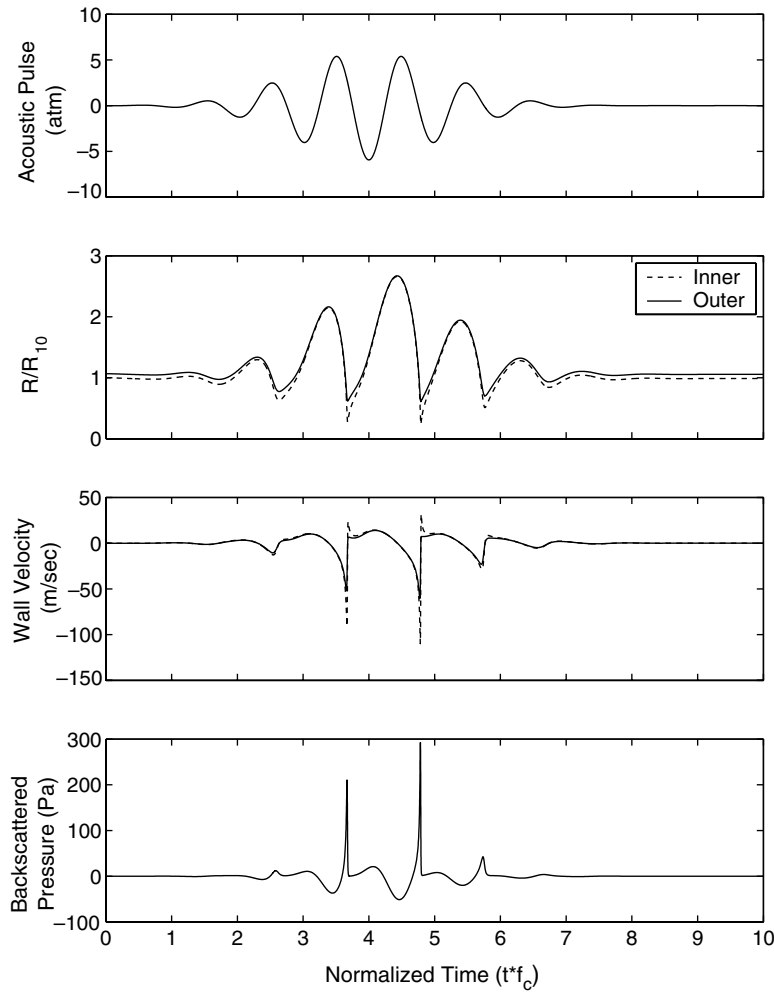


Fig. 2. Bubble response, wall velocity, and backscattered pressure due to an acoustic pulse with mechanical index $MI = 0.4$ ($P_a = 5.93$ atm) and center frequency $f_c = 2.2$ MHz. The bubble ambient radius $R_{10} = 1.5$ μm and the shell thickness is 100 nm.

effect, which decreases wall velocity and pulse echo intensity. This indicates that increasing shell thickness will reduce backscatter. The effect of the shell thickness on the radial motion of a microbubble of inner radius R_{10} ($=1.5$ μm) is shown in Fig. 6 for different shell thicknesses ($=10, 100$, and 500 nm). Lower values of radial expansion and collapse are clearly seen for 500 nm shell thickness than those for thinner shells, but the dependence is weak. Thinner walled bubbles can be expected to have shorter lifetimes when subjected to ultrasound, but for the echo PIV technique, bubble longevity is desired. Thus a tradeoff exists between strong signals from short-lived thin bubbles and lower signal levels from longer-lasting thick-shelled bubbles.

The effect of mechanical index on non-linear dynamics is demonstrated in Fig. 7 for an incident wave frequency f_c ($=2.2$ MHz). Fig. 7 shows the spectral components of the backscattered pressure for different values of MI for a bubble radius of 1.5 μm . At low mechanical index ($MI = 0.2$; $P_a = 2.97$ atm), the mi-

crobubbles produce very weak scatter at both fundamental and second harmonics, with both having similar amplitudes. As the mechanical index increases, at for example $MI = 0.7$ ($P_a = 10.4$ atm), the resonant (peak) frequency for the second harmonic component is lower than the doubled f_c that would be expected for linear bubble dynamics, i.e., $f_2 < 2f_c$. This resonant frequency shift is a well-known non-linear phenomenon, where “detuning” of 5–10% of the expected frequency is seen [36]. This indicates that second harmonic imaging where exactly double the incident center frequency is extracted from the backscattered signal may not be optimal for echo PIV since the maximum backscattered harmonic component will not be detected. Instead, it may be preferable to use a larger bandwidth reception process, centered around the second harmonic but capable of detecting peak amplitudes away from the second harmonic frequency. Also, the shift in harmonic resonance frequency may provide additional information about bubble characteristics; this requires further exploration.

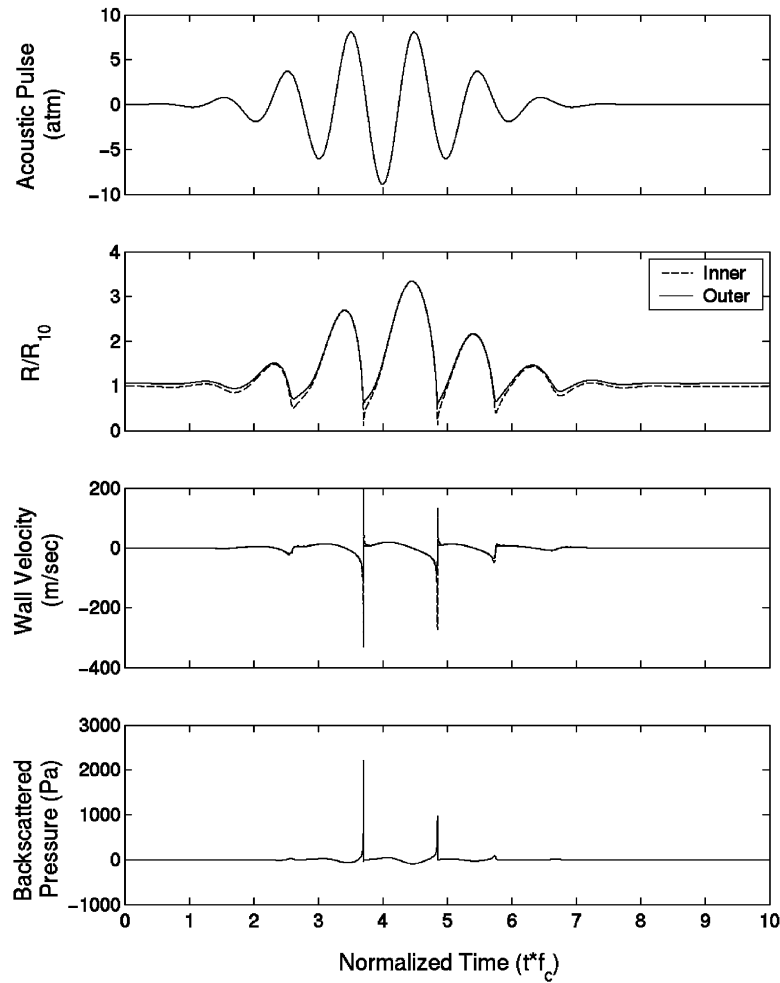


Fig. 3. Bubble response, wall velocity, and backscattered pressure due to an acoustic pulse with mechanical index $MI = 0.6$ ($P_a = 8.90$ atm) and center frequency $f_c = 2.2$ MHz. Bubble dimensions are the same as in Fig. 2.

The effect of varying mechanical index and bubble diameter on the peak amplitude of the second harmonic component is shown in Fig. 8. In this figure, a wide range of bubble sizes with inner radius R_{10} ranging from 0.5 to 5 μm and various mechanical indexes ranging from $MI = 0.1$ – 0.6 ($P_a = 1.48$ – 8.90 atm) are considered. For an incident center frequency of 2.2 MHz, the resonance of the second harmonic is maximized for bubbles with a diameter of approximately 2.75 μm . The amplitude increases monotonically with MI . To excite bubbles with diameter closer to the average available in Optison[®], 2–4.5 μm , lower excitation frequencies should be used. However, lower frequencies can lead to degraded spatial resolution in the resultant B-mode image, although penetration depths may be improved, particularly in large adults. Hence the optimum frequency is coupled to the bubble diameter, but the exact choice depends on the clinical application and the available UCA.

The effect of increasing the incident center frequency is explored in Fig. 9(a) and (b), which shows the ratio of the peak of the second harmonic component to the peak

of the fundamental component for incident pulse frequencies of 2.2 and 6 MHz. Fig. 9(a) shows this ratio for a 2.2 MHz driving pulse. Contour lines show that the scattering from the second harmonic will exceed that of the fundamental for bubbles with inner radius $R_{10} < 1.5$ μm . Hence, there will be no penalty in signal strength for using the second harmonic at these conditions. At the same time, scattering from tissue in the second harmonic is expected to be extremely low. In fact, Duck reports that incident signal amplitudes needed to be increased by 10–25 dB in order for tissue non-linearities to produce any significant second (native) harmonic component [37]. Even with such increases in incident amplitudes, the second harmonic component reached only $\sim 50\%$ of the amplitude of the fundamental. Our results show that even at small values of MI , the second harmonic amplitude of the bubble backscatter can exceed that of the fundamental signal amplitude for resonant bubbles. This lends support to the use of the second harmonic backscatter signal for microbubble detection and PIV analysis.

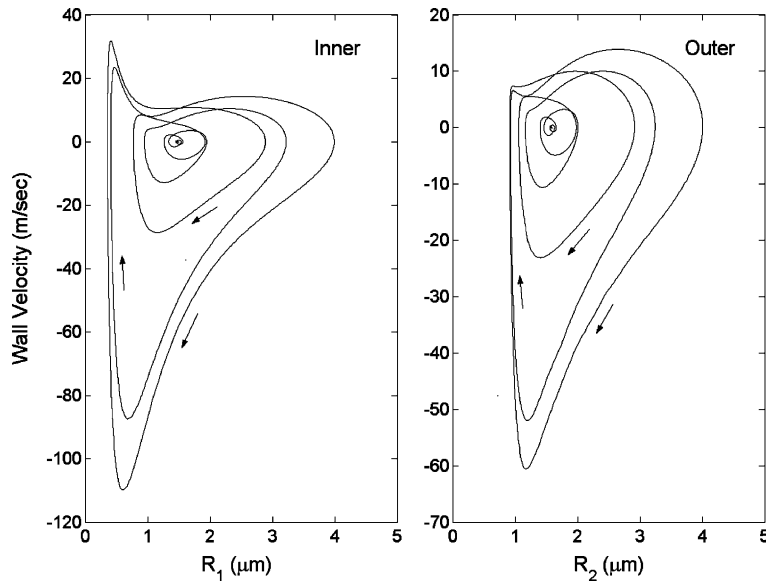


Fig. 4. Phase plot of the wall velocity vs. radial deformation of the inner and outer bubble radii for an incident wave with $MI = 0.4$ ($P_a = 5.93$ atm) and $f_c = 2.2$ MHz. Bubble dimensions are the same as in Fig. 2.

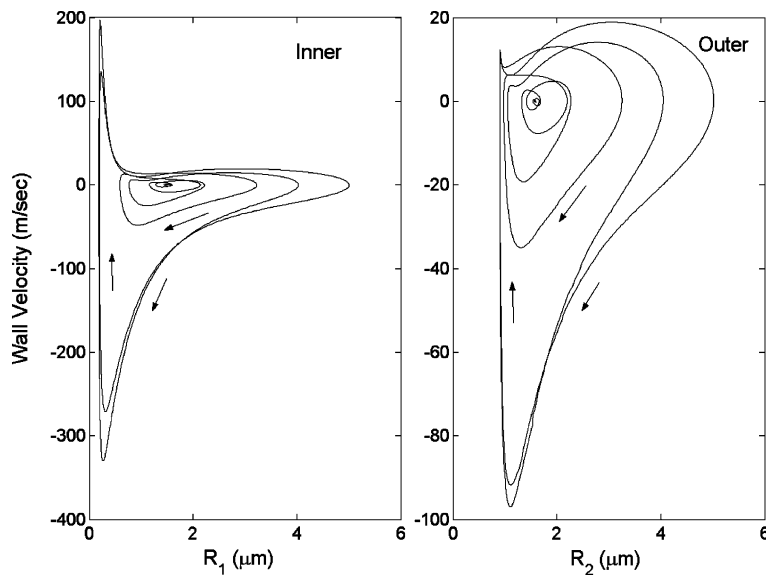


Fig. 5. Phase plot of the wall velocity vs. radial deformation of the inner and outer bubble radii for an incident wave with $MI = 0.6$ ($P_a = 8.90$ atm) and $f_c = 2.2$ MHz. Bubble dimensions are the same as in Fig. 2.

Fig. 9(a) also shows an optimal range of mechanical index ($0.25 < MI < 0.5$) for enhancement of the second harmonic component. Thus, increasing the mechanical index beyond this range will not improve the ratio of the second harmonic to the fundamental component. This implies that echo PIV may be optimized at MI (< 0.6), which can be important in clinical applications where MI must be minimized for safety. In addition, high values of MI lead to UCA destruction [18,24,32]; a low MI setting will extend UCA lifetime, allowing lower dosages to be used, which in turn will reduce the impact of UCA on patients. A similar plot is seen in Fig. 9(b)

for an incident pulse center frequency of 6 MHz, but the enhancement of the second harmonic is shifted toward even smaller bubbles. This again illustrates an optimal range of the incident pulse frequency for the second harmonic technique for a given bubble diameter.

Another factor impacting the response of the bubble is the incident pulse bandwidth. Fig. 10 plots the harmonic response of the fundamental and second harmonic scattering from a microbubble with a radius of $R_{10} = 1.5$ μm and shell thickness 100 nm for different incident pulse bandwidths h_d ($= 0.1, 0.2, 0.3$ and 0.4). For a narrow bandwidth pulse h_d ($= 0.1$), high

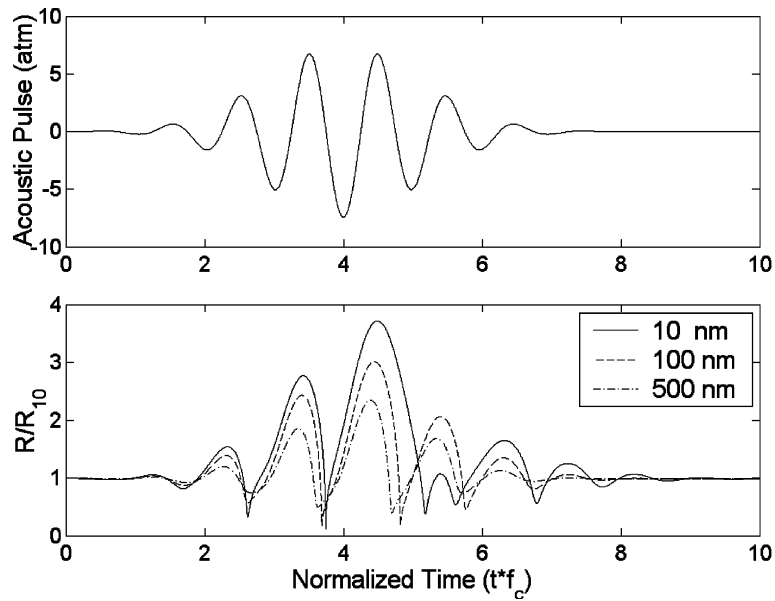


Fig. 6. Comparison of radial bubble response to an acoustic pulse with mechanical index $MI = 0.4$ ($P_a = 5.93$ atm) and center frequency $f_c = 2.2$ MHz. The bubble ambient radius $R_{10} = 1.5$ μm and shell thicknesses ($= 10, 100$, and 500 nm).

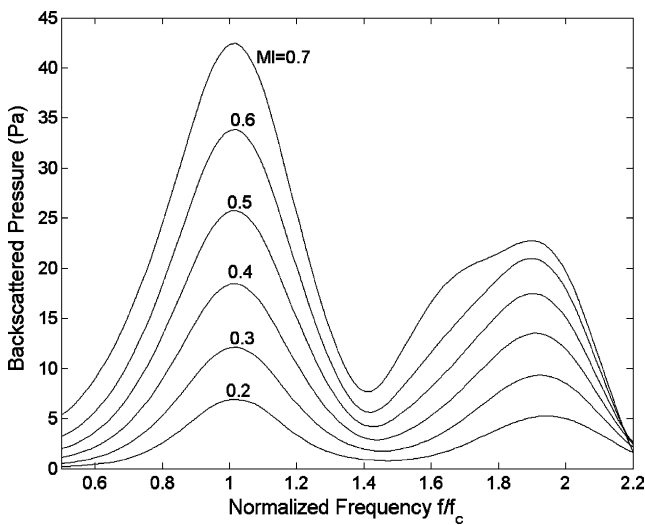


Fig. 7. The spectrum of the fundamental and second harmonic backscattered pressure from a microbubble with $R_{10} = 1.5$ μm and shell thickness of 100 nm for different MI ($= 0.2, 0.3, 0.4, 0.5, 0.6$, and 0.7), and $f_c = 2.2$ MHz. The corresponding incident pressure is: $P_a = MI\sqrt{f_c}$.

scattering intensity is observed for both the fundamental and the second harmonic responses. On the other hand, for broadband pulses $h_d (= 0.4)$, the scattered intensity shows a broad frequency response albeit with lower amplitudes. The effect of the pulse bandwidth on lowering the backscatter intensity agrees with the experimental results reported by Silverman et al. [38]. For implementation of the echo PIV technique, a broadband input pulse with high frequency may be needed to maximize spatial resolution. However, increasing input pulse bandwidth characteristics too much will result in

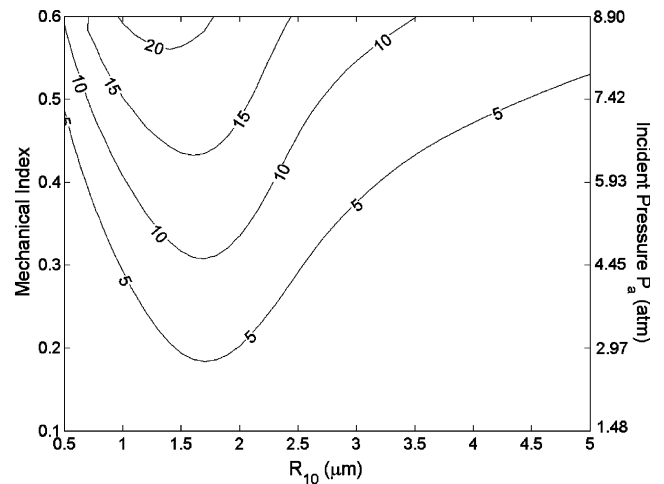


Fig. 8. Contour plot of the second harmonic pressure component (P_a) computed 50 mm away from the center of the microbubble. The shell thickness is 100 nm, and $f_c = 2.2$ MHz.

poorer second harmonic response as shown by Fig. 10. Therefore, a trade-off between maximizing bubble detectability and maximizing spatial resolution will most likely be needed.

Other methods to extract multiple velocity components non-invasively using ultrasound have been proposed. One such method uses multiple transducers and Doppler imaging to reconstruct the velocity vector [39,40]. However, this method has several limitations, some of which we briefly outline here. First, the fundamental nature of Doppler measurement of velocity means that velocity is only measured along the ultrasound beam-line. Using multiple transducer/receiver

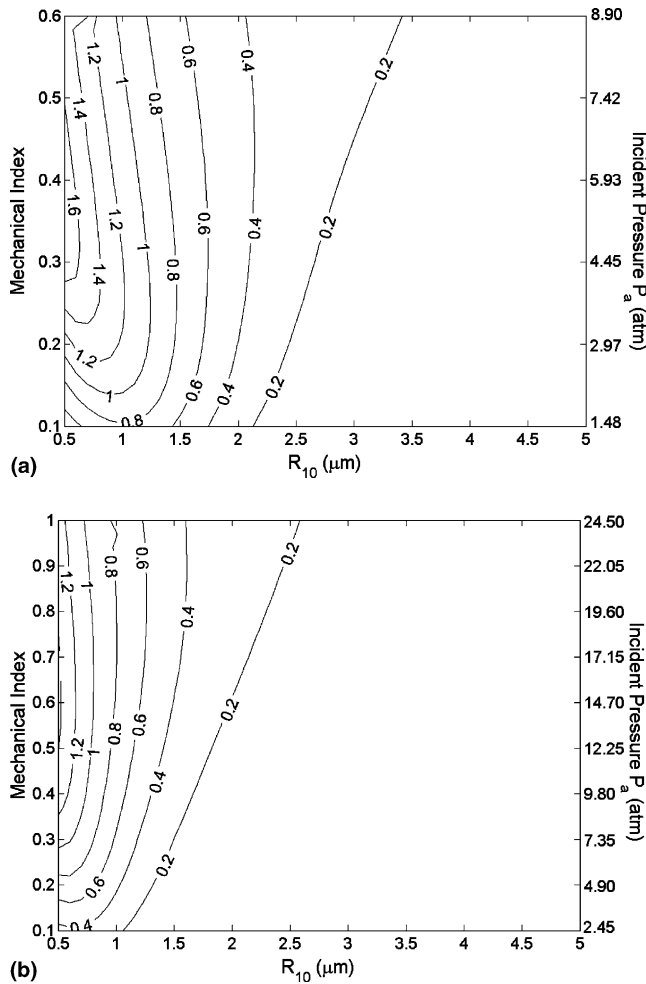


Fig. 9. (a) Contour plot of the ratio of the second harmonic to the fundamental pressure components calculated 50 mm away from the center of the microbubble for different bubble sizes and various values of mechanical index. The shell thickness is maintained at 100 nm, and $f_c = 2.2$ MHz. (b) Contour plot of the ratio of the second harmonic to the fundamental pressure components calculated 50 mm away from the center of the microbubble for different bubble sizes and various values of mechanical index. The shell thickness is 100 nm, and $f_c = 6.0$ MHz.

configurations does not change this. Therefore, these techniques have to make an assumption about the local velocity vector in order to reconstruct the actual velocity component. For 1D (plug-type) flow, this is not an issue since the flow directions is only axial. However, very few flows in the cardiovascular system can be classified as plug flow. Even arterial or venous flows contain secondary and tertiary velocity components. For support, we cite publications by our group [41,42] and MRI velocimetry measurements [43] that show very clearly that flow within various blood vessels can be highly 3-dimensional. For flows at branches, within cardiac chambers, across prosthetic heart valves, across valvar lesions, and within stents and grafts, the situation is highly complex. As an example, consider a vortical flow.

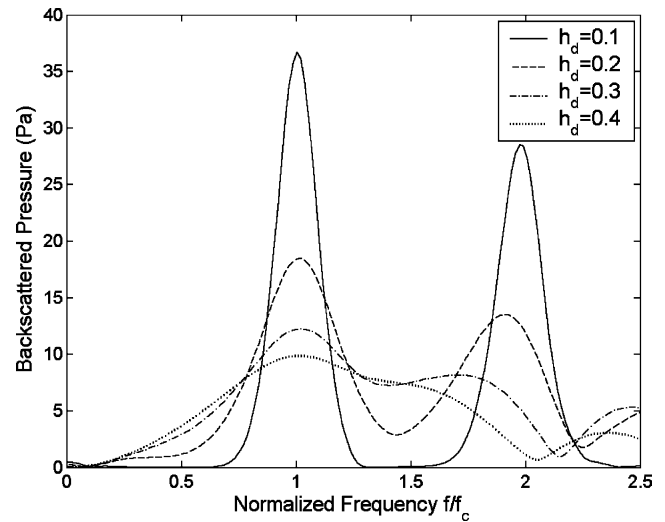


Fig. 10. Pulse bandwidth effect h_d ($=0.1, 0.2, 0.3$, and 0.4) on the backscattered pressure from a microbubble with $R_{10} = 1.5$ μm and shell thickness = 100 nm. $MI = 0.4$ ($P_a = 5.93$ atm), and $f_c = 2.2$ MHz.

To fully reconstruct all velocity components using Doppler techniques, a full 180° of interrogation would be needed, something that is obviously impossible in the clinical setting. Second, it is very difficult to obtain the optimal alignment angle for Doppler measurement of flow in a variety of situations (ex: cardiac jets, shunt flows, pulmonary ductus arteriosus, ventricular septal defects, patent foramen ovale, etc.). Having a larger transducer (such as matrix array transducers) or multiple transducers exacerbates this problem. Even for arterial flow measurements, it is impossible to obtain a truly parallel angle, and either the flow direction is assumed (as for the 2D Doppler array reconstruction techniques) or it is ignored. Frequently, vascular investigators use 60° as the default angle for vascular Doppler—this can produce large errors if the flow angle is significantly different, as can be the case for many arterial geometries. Third, we are not aware of any such method used routinely in the clinical situation. The many advantages inherent in the ability to obtain multi-component velocity vector data would make such a development highly fruitful for any method that can truly make such measurements. It is thus reasonable to question why Doppler-based multi-velocity measurement methods have not developed into commercially usable systems. Fourth, Doppler will still not be able to distinguish between flow and tissue backscatter signals at the endothelial boundary of blood vessels due to the strong backscatter from the tissue interface. Such a capability would be particularly useful for increasing the accuracy of in vivo wall shear stress measurements, something that is done very poorly at present. Lastly, it is important to note that multi-component Doppler methods have had significant problems with reconstructing temporally varying flow.

Other than Doppler, the only ultrasound-based competing technology to that proposed here is ultrasound speckle velocimetry (USV). USV has been in development since the late 1980's [44] and is an ultrasound-equivalent of optical speckle velocimetry techniques that developed from optical interferometry used to study material deformations [45]. USV has shown some promise in the measurement of velocity vectors in arterial flow, but its inherently poor signal to noise ratio, limited ability to resolve flows with high gradients (such as in boundary layers, near walls, etc.), and need for sophisticated transducer configurations have kept this method from being developed into a routine clinical imaging modality. It is interesting to note that several authors including Shung and Flenniken have suggested the use of ultrasound contrast to increase the signal to noise ratio for USV methods [46]. We believe that if contrast is to be used, it makes more sense to use direct particle-based velocimetry techniques such as the echo PIV method rather than an indirect method such as USV, if only because the lower particle densities needed will reduce adverse effects in patients. It is also interesting to note that experimenters in the optical fluid diagnostics area now prefer particle-based techniques over speckle techniques for most velocimetry applications [47,48].

5. Conclusions and future work

This study provides theoretical background for the use of particle-based velocimetry methods to non-invasively measure multiple velocity vectors in vivo using ultrasound. Results show that the non-linear dynamics of small gas-filled microbubbles (ultrasound contrast) provide a means to enhance particle visibility, a prime requirement for PIV processing. Bubble non-linearities at mechanical index (>0.2) induced a resonant frequency shift away from the integer multiple of the incident frequency in the second harmonic backscatter. Results show that the true peak amplitude, which occurs away from an integer multiple of the fundamental frequency, is approximately 30% higher than the amplitude exactly at twice the fundamental frequency. Also, for a given bubble size, there is a combination of optimal incident center frequency and mechanical index range that maximizes the ratio of the second harmonic amplitude compared to the fundamental. This frequency decreases with increasing bubble radius. Further, a narrow bandwidth pulse is shown to increase signal strength. Both of these effects may cause conflict with factors governing spatial resolution, and as with current Doppler imaging, optimization of the incident frequency, microbubble size and mechanical index will depend on the particular clinical application. These theoretical predictions provide a means for initial guidance on the design of an echo PIV system.

Ongoing numerical work ongoing in our laboratory involves further examination of the physics of this process, specifically on the interaction between tissue and bubble backscatter. Small non-linearities in tissue properties may create harmonic signals, especially for deeper targets, which may interfere with the ability to distinguish bubble signatures. An examination of the ideal bubble characteristics that will maximize PIV processing is also underway. Other investigations involve the impact of the incident acoustic signal on bubble translation including acoustic streaming. Ongoing experimental work involves examination of how imaging settings (probe, frequency, frame rate, etc.) in current-generation ultrasound imaging systems can be optimized to maximize PIV accuracy for various cardiovascular applications. Other work includes the development of a first-generation hardware system for real-time echo PIV imaging.

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